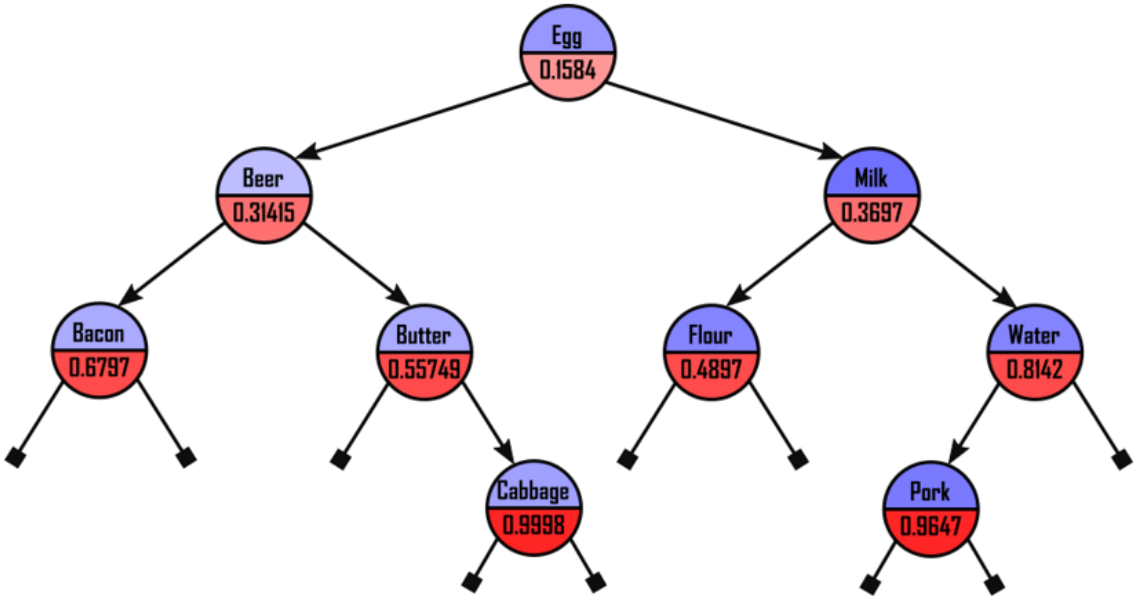


Figure 3.15 An example of a randomized treap (RT) for the keys in the treap shown in figure 3.9 (after inserting the key "Beer"). Priorities are random numbers between 0 and 1. This is just one possible structure for the keys, corresponding to one random choice of the priorities.

Anyway, in the original work by Aragon and Raimund, treaps were introduced as a mean to obtain "randomized balanced search trees"; they used the same idea we described in section 3.4.1, leveraging priorities to force a balanced tree structure, but avoided all the complexity of manually setting these values by using a uniform random numbers generator to choose the values for the nodes' priorities.

Listing 3.9 class RandomizedTreap

```
class RandomizedTreap
  #type Treap
  treap

  #type RandomNumberGenerator
  randomGenerator
```